

CS 229, Fall 2018 Problem set #1 - Supervised Learning

Q 1) Linear classifiers (logistic Regression & GDA).

$$J(\theta) = -\frac{1}{m} \sum_{i=1}^m y^{(i)} \log(h_{\theta}(x^{(i)})) + (1 - y^{(i)}) \log(1 - h_{\theta}(x^{(i)}))$$

$$y^{(i)} \in \{0, 1\}, \quad h_{\theta}(x) = g(\theta^T x) \quad \text{and} \quad g(z) = \frac{1}{1 + e^{-z}}$$

$$\frac{\partial J(\theta)}{\partial \theta} = -\frac{1}{m} \sum_{i=1}^m y^{(i)} \frac{1}{g(\theta^T x^{(i)})} \frac{\partial g(\theta^T x^{(i)})}{\partial \theta} +$$

$$(1 - y^{(i)}) \frac{1}{1 - g(\theta^T x^{(i)})} \frac{\partial g(\theta^T x^{(i)})}{\partial \theta}$$

$$\frac{\partial J(\theta)}{\partial \theta_j} = -\frac{1}{m} \left(\sum_{i=1}^m y^{(i)} \frac{1}{g(\theta^T x^{(i)})} g(\theta^T x^{(i)}) (1 - g(\theta^T x^{(i)})) x_j^{(i)} \right)$$

$$+ (1 - y^{(i)}) \frac{1}{1 - g(\theta^T x^{(i)})} g(\theta^T x^{(i)}) x_j^{(i)}$$

$$(1 - g(\theta^T x^{(i)})) x_j^{(i)}$$

$$= -\frac{1}{m} \sum_{i=1}^m \left[y^{(i)} (1 - g(\theta^T x^{(i)})) - (1 - y^{(i)}) g(\theta^T x^{(i)}) \right] x_j^{(i)}$$

$$= -\frac{1}{m} \sum_{i=1}^m \left[y^{(i)} - g(\theta^T x^{(i)}) \right] x_j^{(i)}$$

$$\frac{\partial J(\theta)}{\partial \theta_j} = \frac{1}{m} \sum_{i=1}^m \left[g(\theta^T x^{(i)}) - y^{(i)} \right] x_j^{(i)}$$

$$\frac{1}{m} X^T (g(X\theta) - Y)$$

$$\frac{\partial^2 J(\theta)}{\partial \theta_i \partial \theta_j} = \frac{1}{m} \sum_{i=1}^m [g(\theta^T x^{(i)}) (1 - g(\theta^T x^{(i)})) - 0]$$

$$x_i^{(i)} \times x_j^{(i)}$$

$$H = \frac{1}{m} [X^T \cdot g(X\theta) \cdot (1 - g(X\theta))] X$$

$$Z^T H Z \geq 0$$

$$\frac{1}{m} \sum_{i=1}^m \sum_{j=1}^n \sum_{k=1}^n g'(\theta^T x^{(i)}) (1 - g'(\theta^T x^{(i)})) x_j^{(i)} x_k^{(i)}$$

$$= \frac{1}{m} \sum_{i=1}^m g'(\theta^T x^{(i)}) (1 - g'(\theta^T x^{(i)})) [(x^{(i)})^T Z]^T Z \geq 0$$

$$(c) \quad p(y) = \begin{cases} \phi & \text{if } y = 1 \\ 1 - \phi & \text{if } y = 0 \end{cases}$$

$$p(x|y=0) = \frac{1}{(2\pi)^{n/2}} \frac{1}{|\Sigma|^{1/2}} \exp\left(-\frac{1}{2} (x - \mu_0)^T \Sigma^{-1} (x - \mu_0)\right)$$

$$p(x|y=1) = \frac{1}{(2\pi)^{n/2}} \frac{1}{|\Sigma|^{1/2}} \exp\left(-\frac{1}{2} (x - \mu_1)^T \Sigma^{-1} (x - \mu_1)\right)$$

by Bayes's rule

$$P(y=1|x) = \frac{p(x|y=1) \cdot P(y=1)}{P(x)}$$

$$\& \quad P(x) = p(x|y=1) \cdot P(y=1) + p(x|y=0) \cdot P(y=0)$$

$$p(y=1|x) = \frac{\phi}{(2\pi)^{n/2} |\Sigma|^{1/2}} \exp\left(-\frac{1}{2} (x - \mu_0)^T \Sigma^{-1} (x - \mu_0)\right)$$

$$\frac{\phi}{(2\pi)^{n/2} |\Sigma|^{1/2}} \exp\left(-\frac{1}{2} (x - \mu_0)^T \Sigma^{-1} (x - \mu_0)\right) +$$

$$\frac{1-\phi}{(2\pi)^{n/2} |\Sigma|^{1/2}} \exp\left(-\frac{1}{2} (x - \mu_1)^T \Sigma^{-1} (x - \mu_1)\right)$$

$$= \frac{1}{1 + \frac{1-\phi}{\phi} \exp \left(-\frac{1}{2} (x - \mu_1)^T \Sigma^{-1} (x - \mu_1) \right) \exp \left(-\frac{1}{2} (x - \mu_0)^T \Sigma^{-1} (x - \mu_0) \right)}$$

$$= \frac{1}{1 + \ln \left(\frac{\exp \left(-\frac{1}{2} (x - \mu_1)^T \Sigma^{-1} (x - \mu_1) \right) \cdot \ln \left(\frac{1-\phi}{\phi} \right)}{\exp \left(-\frac{1}{2} (x - \mu_0)^T \Sigma^{-1} (x - \mu_0) \right)} \right)}$$

$$= \frac{1}{1 + \exp \left[-\left(\frac{1}{2} (x - \mu_1)^T \Sigma^{-1} (x - \mu_1) + \frac{1}{2} (x - \mu_0)^T \Sigma^{-1} (x - \mu_0) - \ln \left(\frac{1-\phi}{\phi} \right) \right) \right]}$$

$$= \frac{1}{1 + \exp \left(- \left[\Sigma^{-1} (\mu_1 - \mu_0)^T x + \frac{1}{2} (\mu_0 + \mu_1)^T \Sigma^{-1} (\mu_0 - \mu_1) - \ln \left(\frac{1-\phi}{\phi} \right) \right] \right)}$$

$$\theta = \Sigma^{-1} (\mu_1 - \mu_0)^T$$

$$\phi_0 = \frac{1}{2} (\mu_0 + \mu_1)^T \Sigma^{-1} (\mu_0 - \mu_1) - \ln \left(\frac{1-\phi}{\phi} \right)$$

d) $\mu_{y(i)}$

$$\mu_{y(i)} = \mathbb{I}\{y^{(i)} = 0\} \mu_0 + \mathbb{I}\{y^{(i)} = 1\} \mu_1$$

$$p(x^{(i)} | y^{(i)}; \mu_0, \mu_1, \Sigma) = \frac{1}{(2\pi)^{n/2} |\Sigma|^{1/2}} \exp$$

$$p(x^{(i)} | y^{(i)}; \mu_0, \mu_1, \Sigma) =$$

$$\frac{1}{(2\pi)^{n/2} |\Sigma|^{1/2}} \exp \left\{ -\frac{1}{2} (x - \mu_{y(i)})^T \Sigma^{-1} (x - \mu_{y(i)}) \right\}$$

$$p(y^{(i)}; \phi) = \phi^{\mathbb{I}\{y^{(i)}=1\}} \cdot (1-\phi)^{1-\mathbb{I}\{y^{(i)}=1\}}$$

$$l = \sum_{i=1}^M \log(p(x^{(i)} | y^{(i)}; \mu_0, \mu_1, \Sigma) +$$

↑ likelihood

$$\sum_{i=1}^m \log(p(y^{(i)}; \phi))$$

$$l = \sum_{i=1}^m \log \left(\frac{1}{(2\pi)^{n/2} |\Sigma|^{1/2}} \exp \left\{ -\frac{1}{2} (x - \mu_{y(i)})^T \Sigma^{-1} (x - \mu_{y(i)}) \right\} \right. \\ \left. + \sum_{i=1}^m \log \left(\phi^{\mathbb{I}\{y^{(i)}=1\}} \cdot (1-\phi)^{1-\mathbb{I}\{y^{(i)}=1\}} \right) \right)$$

$$l = -\frac{mn}{2} \log(2\pi) - \frac{m}{2} \log |\Sigma| - \frac{1}{2} \sum_{i=1}^m (x^{(i)} - \mu y^{(i)})^T \Sigma^{-1} (x^{(i)} - \mu y^{(i)})$$

$$+ \sum_{i=1}^m I\{y^{(i)} = 1\} \log \phi + (m - \sum_{i=1}^m I\{y^{(i)} = 1\}) \log(1-\phi)$$

$$\frac{\partial l}{\partial \phi} = 0 = \frac{\sum_{i=1}^m I\{y^{(i)} = 1\}}{\phi} + \frac{m - \sum_{i=1}^m I\{y^{(i)} = 1\}}{-(1-\phi)}$$

$$\frac{\sum_{i=1}^m I\{y^{(i)} = 1\}}{\phi} = \frac{\sum_{i=1}^m I\{y^{(i)} = 1\} - m}{1-\phi}$$

$$\frac{1-\phi}{\phi} \sum_{i=1}^m I\{y^{(i)} = 1\} - \sum_{i=1}^m I\{y^{(i)} = 1\} = -m$$

$$\sum_{i=1}^m I\{y^{(i)} = 1\} \left[1 - \frac{1-\phi}{\phi} \right] = m$$

$$\phi = 1$$

$$\frac{\partial l}{\partial \phi} = 0 = \frac{\sum_{i=1}^m I\{y^{(i)} = 1\}}{\phi} + \frac{1}{\phi-1} \left(m - \sum_{i=1}^m I\{y^{(i)} = 1\} \right)$$

$$\text{Let } K = \sum_{i=1}^m I\{y^{(i)} = 1\}$$

$$\frac{\partial \mathcal{L}}{\partial \phi} = \frac{K}{\phi} + \frac{1}{\phi-1} (m-K) = 0$$

$$\frac{K}{\phi} = \frac{K-m}{\phi-1}$$

$$\frac{\phi-1}{\phi} K = K-m$$

$$m = K \left(1 - \frac{\phi-1}{\phi} \right)$$

$$m = K \left(\frac{1}{\phi} \right)$$

$$\phi = \frac{K}{m}$$

$$\phi = \frac{1}{m} \sum_{i=1}^m I\{y^{(i)} = 1\}$$

$$\frac{\partial \mathcal{L}}{\partial \Sigma} = -\frac{m}{2} \frac{1}{\Sigma} + \frac{1}{2} \frac{1}{(\Sigma)^2} \left(\sum_{i=1}^m (x^{(i)} - \mu_{y(i)}) (x^{(i)} - \mu_{y(i)})^T \right)$$

$$\frac{\partial \mathcal{L}}{\partial \Sigma} = 0$$

$$-\frac{m}{2\Sigma} + \frac{1}{2} \frac{1}{(\Sigma)^2} \left(\sum_{i=1}^m (x^{(i)} - \mu_{y(i)}) (x^{(i)} - \mu_{y(i)})^T \right) = 0$$

$$-m + \frac{1}{(\Sigma)} \left(\sum_{i=1}^m (x^{(i)} - \mu_{y(i)}) (x^{(i)} - \mu_{y(i)})^T \right) = 0$$

$$\Sigma = \frac{1}{m} \left(\sum_{i=1}^m (x^{(i)} - \mu_{y(i)}) (x^{(i)} - \mu_{y(i)})^T \right)$$

$$\mu_0 = \sum_{i=1}^m (y^{(i)} = 0) x^{(i)}$$

$$y^{(i)} = 0$$

$$\mu_1 = \sum_{i=1}^m (y^{(i)} = 1) x^{(i)}$$

$$y^{(i)} = 1$$

Q2) a) $p(y^{(i)} = 1 | t^{(i)} = 1, x^{(i)}) = p(y^{(i)} = 1 | t^{(i)} = 1)$

$$\Rightarrow p(y=1 | t=1, x) p(t=1 | x) p(x) = p(y=1, t=1, x) = p(t=1 | y=1, x) \cdot p(y=1 | x)$$

$$p(t=1 | x) = \frac{p(y=1 | x) \cdot p(t=1 | y=1, x)}{p(y=1 | t=1, x)}$$

$$p(t=1 | y=1, x) = 1, \quad p(y=1 | t=1, x) = p(y=1 | t=1)$$

$$p(t=1 | x) = \frac{p(y=1 | x)}{p(y=1 | t=1)}$$

$$p(y=1 | t=1) = \alpha$$

b) Assuming that $h(x^{(i)}) \approx p(y^{(i)}=1 | x^{(i)})$ for all examples $x^{(i)}$

$$h(x) \approx p(y=1 | x) = p(t=1 | x) \alpha \approx \alpha \quad \text{for all } x \in V_X$$

Q3.) Poisson Regression $\rightarrow$

$$p(y; \lambda) = \frac{e^{-\lambda} \lambda^y}{y!}$$

$$p(y; \lambda) = \exp\left(\log\left(\frac{e^{-\lambda} \lambda^y}{y!}\right)\right)$$

$$= \exp(\log e^{-\lambda} + \log \lambda^y - \log y!)$$

$$= \exp(-\lambda \log e + y \log \lambda - \log y!)$$

$$= \exp(-\lambda + y \log \lambda + \log \frac{1}{y!}) \Rightarrow e^{a+b+c} = e^a \cdot e^b \cdot e^c$$

$$= \exp(-\lambda) \cdot \exp(y \log \lambda) \cdot \exp\left(\log \frac{1}{y!}\right)$$

$$= \frac{1}{y!} \exp(-\lambda) \cdot \exp(y \log \lambda)$$

$$= \frac{1}{y!} \exp(y \log \lambda - \lambda)$$

$$\boxed{b(y) = \frac{1}{y!}}$$

$$T(y) = y \quad - \quad a(\eta) = -\lambda$$

$$\boxed{\eta = \log \lambda}$$

$$\begin{aligned} a(\eta) &= \lambda \\ \boxed{a(\eta) &= e^\eta} \end{aligned}$$

Canonical response function $\Rightarrow E(y_i | x_i)$
 mean. $= 1$

$$h_{\theta}(x) = \lambda = e^{\eta}$$

$$h_{\theta}(x) = \lambda = e^{\theta^T x}$$

(c) For all GLM's, the update rule is as follows $\rightarrow$

$$\theta_j = \theta_j + \alpha (y^{(i)} - h_{\theta}(x^{(i)})) x_j^{(i)}$$

$$\theta_j = \theta_j + \alpha (y^{(i)} - e^{\theta^T x^{(i)}}) x_j^{(i)}$$

CS229 Fall 2018, Problem Set #1 - supervised learning

Q4. a) $p(y; \eta) = b(y) \exp(\eta^T T(y) - a(\eta))$

$$\frac{\partial}{\partial \eta} \int p(y; \eta) dy = \int \frac{\partial}{\partial \eta} p(y; \eta) dy$$

$\downarrow$
0

$$= \int \cancel{b(y)} \frac{\partial}{\partial \eta} b(y) \exp(\eta^T T(y) - a(\eta)) dy$$

$$= \int \cancel{b(y) \exp(\eta^T T(y) - a(\eta))}$$

$$= \int \frac{\partial}{\partial \eta} b(y) \exp(\eta^T y - a(\eta)) dy$$

$$= \int b(y) \exp(\eta y - a(\eta)) \left(y - \frac{\partial a(\eta)}{\partial \eta} \right) dy$$

If η is scalar, then $\eta = \eta^T$

$$= \int p(y; \eta) \left(y - \frac{\partial a(\eta)}{\partial \eta} \right) dy$$

$$= \int y p(y; \eta) dy - \frac{\partial a(\eta)}{\partial \eta} \int p(y; \eta) dy$$

$$= E[y; \eta] - \frac{\partial a(\eta)}{\partial \eta} = 0$$

$$\boxed{E[y; \eta] = \frac{\partial a(\eta)}{\partial \eta}}$$

$$(b) \quad \frac{\partial}{\partial \eta} \int y p(y; \eta) dy = \frac{\partial^2 a(\eta)}{\partial \eta^2}$$

$$\frac{\partial}{\partial \eta} \int y p(y; \eta) dy = \int y \frac{\partial}{\partial \eta} p(y; \eta) dy$$

$$= \int y p(y; \eta) \left(y - \frac{\partial a(\eta)}{\partial \eta} \right) dy$$

$$= \int y^2 p(y; \eta) dy - \frac{\partial a(\eta)}{\partial \eta} \int y p(y; \eta) dy$$

$$= E[Y^2; \eta] - E^2[Y; \eta]$$

$$= \text{var}[Y; \eta] = \cancel{\text{var}[Y; \eta]} = \text{var}[Y|X; \theta]$$

CS 229 Fall 2018 Problem Set #1 - Supervised Learning

Q5) Locally Weighted Regression

a) For locally weighted regression, we need to fit θ to minimize,

$$\sum_{i=1}^m w^{(i)} (y^{(i)} - \theta^T x^{(i)})^2$$

$$\Rightarrow$$

$$w^{(1)} (y^{(1)} - \theta^T x^{(1)})^2 + w^{(2)} (y^{(2)} - \theta^T x^{(2)})^2 \dots$$

$$\text{Since, } h_{\theta}(x^{(i)}) = (x^{(i)})^T \theta$$

$$X\theta - \vec{y} = \begin{bmatrix} h_{\theta}(x^{(1)}) - y^{(1)} \\ \vdots \\ h_{\theta}(x^{(m)}) - y^{(m)} \end{bmatrix}$$

$$\text{For a vector } z, \quad z^T z = \sum z_i^2$$

$$\text{Also, } z^T W z \Rightarrow \sum w_i z_i^2$$

$$\downarrow$$

$$m \times m \quad \text{if } z \text{ is } m \times 1$$

$$\text{So } \frac{1}{2} (X\theta - \vec{y})^T W (X\theta - \vec{y}) = \frac{1}{2} \sum_{i=1}^m w^{(i)} (y^{(i)} - \theta^T x^{(i)})^2$$

$$= J(\theta)$$

$$W_{ij} = \begin{cases} \frac{1}{2} w^{(i)} & i=j \\ 0 & i \neq j \end{cases}$$

$$(ii) J(\theta) = (X\theta - y)^T W (X\theta - y)$$

$$J(\theta) = (\theta^T X^T - y^T) W (X\theta - y)$$

$$J(\theta) = (\theta^T X^T W - y^T W) (X\theta - y)$$

$$J(\theta) = (\theta^T X^T W X\theta - \theta^T X^T W y - y^T W X\theta - y^T W y)$$

↓
Taking trace both sides, since we know $J(\theta)$ is a real number, trace of real number is a real number.

$$\text{tr } J(\theta) = \text{tr} ()$$

$$\cancel{\text{tr } J(\theta)} \quad \text{Also, } \text{tr } A = \text{tr } A^T$$

$$\text{so } \text{tr } \theta^T X^T W y = \text{tr } y^T W X \theta$$

$$\text{tr } J(\theta) = \text{tr} (\theta^T X^T W X\theta) - 2 \text{tr} (y^T W X\theta) - \text{tr} (y^T W y)$$

↓

Taking derivative wrt θ

$$\boxed{\nabla_{\theta} \text{tr} (y^T W y) = 0}$$

$$\nabla_{\theta} \text{tr } J(\theta) = 0 = \text{tr} [\theta^T X^T W X] - 2 [y^T W X]$$

↑
to minimize $J(\theta)$

we know

$$\nabla_A \text{tr} A B A^T C = C A B + C^T A B^T$$

$$A = \theta^T$$

$$B = X^T W X$$

$$C = I$$

$$\nabla_{\theta} \text{tr} J(\theta) = \theta^T X^T W X + \theta^T X^T W X - 2 (y^T W X) = 0$$

$$2(\theta^T X^T W X) - 2(y^T W^T X) = 0$$

$$\theta^T X^T W X = y^T W^T X$$

↓ take transpose both sides

$$X^T W^T X \theta = X^T W^T y$$

$$\theta = (X^T W^T X)^{-1} (X^T W y)$$